INFO 4220 – Networks II: Market Design Spring 2025 – Problem Set 1

SOLUTION

Problem 1 (12 points, 3 pts each)

The supply curve for natural gas per thousand cubic feet (per Tcf) in the US is given by $Q_S = 18.5 + 0.7 * P_G$ and the demand curve for natural gas is $Q_D = 34.5 - 1.8 * P_G$.

a) Find the market equilibrium for natural gas

Answer:

We set $Q_d = Q_s \Rightarrow 18.5 + 0.7 * P_G = 34.5 - 1.8 * P_G \Rightarrow P_G = 6.4, Q_S = Q_D = 22.98$

b) Assume that the US government passes a law that sets a ceiling for the price of natural gas at $7 per Tcf. What is the effect of the price ceiling on the market equilibrium you found in (a)

Answer: The price cap has no effect as the equilibrium price is lower. (2pt for conclusion and 1pt for explanation.)

c) Environmental groups campaign and manage to get the government to set a maximum production quota of 20 Tcf. What is the effect of this quota on the market equilibrium you found in (a)

Answer:

If the quota is set at 20, then the price will be given by $20 = 34.5 - 1.8 * P_G \Rightarrow P_G = 8.06$

(2pt for the formular and 1pt for the final price)
(partial credit if calculating the price with the supply curve: in this case, the price will be determined by the demand side.)

d) Which scenario (a, b, or c) will be better for producers and for consumers.

Answer: We need to compare the consumer and producer surplus in each part.

For part (a):
CS = $((19.17 - 6.4) * 22.98)/2 = 146.69$
$PS = 6.4 * 18.5 + ((22.98 - 18.5) *$6.4)/2=132.74
(1pt)

For part (c):

$CS = (19.17 - 8.06) * \frac{20}{2} = 111.11$

$PS = 8.06 * 20 - \frac{1.5*2.14}{2} = 159.51$

(1pt)

Consumers prefer part (a), producers prefer part (c) (0.5pt for each side)

Problem 2 (12 points, 3 pts each)

Ella and Sam, a couple, are deciding where to move. They have four options: **Apartment A, Apartment B, Apartment C,** or **Staying at their current place**. Ella strongly prefers **Apartment A**, while Sam's top choice is **Apartment B**. Both want to avoid conflict, so they will independently write down their decision and reveal it at the same time.

The payoffs for each choice are given in the matrix below (known to both Ella and Sam):

		Sam			
		Apt A	**Apt B**	**Apt C**	**Stay**
Ella	**Apt A**	(50, 20)	(-20,-20)	(10, 30)	(0, 0)
	Apt B	(-20, -20)	(30, 50)	(5, 10)	(20, -10)
	Apt C	(10, 5)	(5, 10)	(20, 20)	(0, -5)
	Stay	(0, 0)	(0, -10)	(0, 5)	(10, 10)

a) Iteratively delete all dominated strategies. Show the resulting payoff matrix.

Answer:
For Sam, Apt A is dominated by Apt C, we can delete Sam - Apt A
Now for Ella, Stay is dominated by Apt B, we can delete Ella – Stay
Now for Sam, Stay is dominated by by Apt C, we can delete Sam – Stay
Now for Ella, Apt A is dominated by Apt C, we can delete Ella – Apt A

(0.75pt for each correct deletion)
We are left with the following matrix:

		Sam			
		~~**Apt A**~~	**Apt B**	**Apt C**	~~**Stay**~~
Ella	~~**Apt A**~~	~~(50, 20)~~	~~(-20,-20)~~	~~(10, 30)~~	~~(0, 0)~~
	Apt B	~~(-20, -20)~~	(30, 50)	(5, 10)	~~(20, -10)~~
	Apt C	~~(10, 5)~~	(5, 10)	(20, 20)	~~(0, -5)~~
	~~**Stay**~~	~~(0, 0)~~	~~(0, -10)~~	~~(0, 5)~~	~~(10, 10)~~

b) Is there a Nash Equilibrium in **pure strategies**? If yes, find all of them and prove that it satisfies the definition of a Nash Equilibrium.

Answer:

There are 2 N.E: (Apt B, Apt B) and (Apt C, Apt C) (1.5pt for each)

c) Are there any strategy profiles that are **Pareto efficient**? Discuss whether you expect any of these to be the outcome of the game.

Answer:

While there are 2 N.E., (Apt C, Apt C) is not pareto efficient because in (Apt B, Apt B) both players will get a higher payoff. (Apt B, Apt B) are pareto efficient. (2pt)
I would expect (Apt B, Apt B) to be the outcome of the game. (1pt)

d) True/False (explain why): A weakly dominated strategy for player *i* can never be part of a Nash Equilibrium

Answer:

False! A weakly dominated strategy can be part of a N.E. (2pt for conclusion, 1pt for explanation)

Problem 3 (14 points)

Glinda and Elphaba must decide how to act in Oz amidst rising tensions between Wizard supporters and the Animals' rights movement. Elphaba can either **confront the Wizard directly (CD)** or **work quietly behind the scenes (WS)** to avoid attracting attention. Glinda can either **support the Wizard (SW)** to maintain her public image or **secretly assist Elphaba (SE)** in helping the Animals. Their choices impact their outcomes based on the state of public opinion in Oz:

- **Pro-Wizard (PW)**: Majority support the Wizard (Probability 2/3)
- **Pro-Animal (PA)**: Growing support for the Animals (Probability 1/3)

Glinda and Elphaba will work separately, and they will not be able to communicate with each other during this time. **Only Elphaba will receive the signal of public opinion** with the help of flying monkeys, who reliably report whether the true state of the world is PW or PA. Glinda does not receive this signal but knows the prior probabilities of this signal. The payoffs are shown in the payoff matrix below:

Pro-Wizard (PW)				**Pro-Animal (PA)**			
		Elphaba				**Elphaba**	
		Confront Wizard (CD)	**Work Quietly (WQ)**			**Confront Wizard (CD)**	**Work Quietly (WQ)**
Glinda	**Support Wizard (SW)**	(30, -50)	(40, 10)	**Glinda**	**Support Wizard (SW)**	(-20, 50)	(10, 40)
	Secretly Assist (SA)	(-10, -20)	(0, 0)		**Secretly Assist (SA)**	(50, 100)	(40, 30)

a) Explain what defines a Bayesian Nash Equilibrium in this game (6 points).

Answer:

A B.N.E in this game is a set of strategy profile *s* that satisfy:

For each player $i \in \{E, G\}$ and each signal they receive t_i, and for each action $a_i \in A_i$the expected payoff of player i must be higher when playing the equilibrium strategy than when playing a different strategy.
(get partial credits if only showing the def of BNE)

In other words, for Elphaba, it will be a strategy profile S_E that is based on the signal she observes: what she would do if the state of Oz is Pro-Wizard (PW), and what she would do if the state of Oz is Pro-Animal (PA). (3pts)

Optional for answering this in (a), could leave it for (b): When the state of Oz is Pro-Wizard (PW), working quietly (WQ) dominates confronting the wizard directly (CD): she gets a payoff of 10 or 0 (depending on whether Glinda SW or SA) if she WQ, and -50 or -20 if she CD. When the state of Oz is Pro-Animal (PA), confronting the wizard directly (CD) dominates working quietly (WQ).

So Elphaba's strategy S_E is: working quietly (WQ) if Pro-Wizard (PW), confronting the wizard directly (CD) if Pro-Animal (PA).

For Glinda, she does not get the signal of the state of Oz, but she gets the probability of the state of Oz (and therefore Elpahaba's strategy). We can write her strategy profile SA as (G's action, E's strategy): (G's action, (E's action if PW, E's action if PA)) (3pts)

b) Find the Bayesian Nash Equilibrium for this game. For doing so, list all possible strategy profiles and compute the expected payoff for each. Then determine which strategy profiles constitute a B.N.E. (8 points).

Answer:

Strategy	G's expected payoff	E's payoff if PW	E's payoff if PA
(SW, (CD, CD))	⅔ * 30 + ⅓ * (-20) = 40/3	-50	50
(SW, (CD, WQ))	⅔ * 30 + ⅓ * 10 = 70/3	-50	40
(SW, (WQ, CD))	⅔ * 40 + ⅓ * (-20) = 20	10	50
(SW, (WQ, WQ))	⅔ * 40 + ⅓ * 10 = 30	10	40
(SA, (CD, CD))	⅔ * (-10) + ⅓ * 50 = 10	-20	100
(SA, (CD, WQ))	⅔ * (-10) + ⅓ * 40 = 20/3	-20	30
(SA, (WQ, CD))	⅔ * 0 + ⅓ * 50 = 50/3	0	100
(SA, (WQ, WQ))	⅔ * 0 + ⅓* 40 = 40/3	0	30

1pt for listing all the 8 combinations, 1 pt for G's expected payoff, 1pt for E's payoff if PW, 1pt for E's payoff if PA. Deduct –0.25 for each incorrect answer.

The best response of Elphaba when Glinda chooses to support wizard (SW) is the strategy profile (WQ, CD). Elphaba could get 10 if PW, and 50 if PA. When Glinda chooses to secretly assist (SA), the best response for Elphaba will be (WQ. CD). That being said, regardless of Glinda's action, Elphaba's best strategy will be (WQ, CD): WQ if PW, CD if PA. (1pt for E's best strategy)

When Elphaba always chooses (WQ, CD), the best response for Glinda will be supporting wizard (SW), since uG(SW, (WQ, CD))=20 >uG(SA, (WQ, CD)) = 50/3 (1pt for G's best strategy)

And the BNE for this game is (SW, (WQ, CD)). (2pt for the correct BNE.)

Problem 4 (12 points)

Five children—Alice, Ben, Clara, Daniel, and Emma—are deciding which of five dogs to adopt. Dogs also have preferences for the children they'd like to live with. Using the Deferred Acceptance Algorithm, match each child to a dog they will adopt.

Children's Preferences

Alice	Ben	Clara	Daniel	Emma
Bella	Rocky	Luna	Max	Bella
Luna	Max	Max	Charly	Luna
Max	Bella	Rocky	Rocky	Charly
Charly	Luna	Bella	Luna	Max
Rocky	Charly	Charly	Bella	Rocky

Dog's Preferences

Max	Bella	Rocky	Luna	Charly
Daniel	Emma	Ben	Clara	Alice
Ben	Alice	Clara	Emma	Emma
Clara	Clara	Daniel	Alice	Daniel
Emma	Ben	Emma	Ben	Ben
Alice	Daniel	Alice	Daniel	Clara

a) Using the Deferred Acceptance Algorithm, where the children propose and the dogs accept or reject proposals, determine the final stable matching. Show all the steps and explain your work. (4 points)

Answer:

Step 1: Initial Proposals (1pt)

- **Alice:** Proposes to Bella. Bella tentatively accepts Alice.
- **Ben:** Proposes to Rocky. Rocky tentatively accepts Ben.
- **Clara:** Proposes to Luna. Luna tentatively accepts Clara.
- **Daniel:** Proposes to Max. Max tentatively accepts Daniel.
- **Emma:** Proposes to Bella. Bella rejects Alice and Accepts Emma.

Tentative Matching After Step 1:

- Bella → Emma
- Rocky → Ben
- Luna → Clara
- Max → Daniel
- Charlie → Unmatched
- Alice → Unmatched

Step 2: Next Round of Proposals

1) **Alice:** Proposes to Luna. Luna rejects Alice because Luna prefers Clara. (1pt)
2) **Alice:** Proposes to Max. Max rejects Alice, because Max prefers Daniel. (1pt)
3) **Alice:** Proposes to Charlie. Charlie tentatively accepts Alice. (1pt)

Tentative Matching After Step 2:

- Bella → Emma
- Rocky → Ben
- Luna → Clara
- Max → Daniel
- Charlie → Alice

Final Stable Matching (Children Proposing):

- **Bella:** Emma
- **Rocky:** Ben
- **Luna:** Clara
- **Max:** Daniel
- **Charlie:** Alice

(Partial credit for listing the correct final matching without showing the steps.)

b) A sixth child arrives before any decision is made, and a new dog is introduced to the options, changing everyone's preferences. However, Clara is in a rush and pays an extra fee to be matched before the rest. However, her match will be random, with an equal probability of getting each dog.

In practice, what happens is that first Clara gets a random dog, and then the other 5 kids are matched to the remaining dogs using D.A. with kids proposing. What is the likelihood that this process leads to a non-stable matching? Below are the new preferences you should use for this part. Show and explain all your work (8 points)

Alice	**Ben**	**Clara**	**Daniel**	**Emma**	**Frank**
Bella	Rocky	Luna	Max	Bella	Charlie
Max	Max	Max	Daisy	Luna	Daisy
Luna	Bella	Daisy	Charlie	Charlie	Rocky
Daisy	Charlie	Rocky	Rocky	Max	Max
Rocky	Daisy	Bella	Bella	Rocky	Luna
Charlie	Luna	Charlie	Luna	Daisy	Bella

Max	**Bella**	**Rocky**	**Luna**	**Charlie**	**Daisy**
Clara	Emma	Ben	Clara	Frank	Clara
Daniel	Alice	Clara	Emma	Alice	Frank
Ben	Clara	Daniel	Ben	Daniel	Daniel
Alice	Ben	Emma	Alice	Emma	Emma
Emma	Frank	Frank	Frank	Ben	Ben
Frank	Daniel	Alice	Daniel	Clara	Alice

Answer:

There are 5 matches that is not stable since there is a blocking pair:
For example, **three cases** where Clara is assigned a random dog and identify **blocking pairs** based on the new preferences.

Case 1: Clara → Max

1. **Initial Forced Match**: Clara → Max.
2. **Final Matching (Children-Proposing Algorithm)**:
    - Alice → Bella
    - Ben → Rocky
    - Clara → Max
    - Daniel → Daisy
    - Emma → Charlie
    - Frank → Luna

Blocking Pair(s):

- **Clara and Luna**:
    - Clara prefers Luna (ranked 1st) over Max (3rd).
    - Luna prefers Clara (ranked 1st) over Frank (6th).
    - **Blocking pair: (Clara, Luna).**

Case 2: Clara → Bella

1. **Initial Forced Match**: Clara → Bella.
2. **Final Matching (Children-Proposing Algorithm)**:
    - Alice → Max
    - Ben → Rocky
    - Clara → Bella

- Daniel → Daisy
- Emma → Charlie
- Frank → Luna

Blocking Pair(s):

- **Alice and Bella**:
  - Alice prefers Bella (ranked 1st) over Max (2nd).
  - Bella prefers Alice (ranked 2nd) over Clara (3rd).
  - **Blocking pair: (Alice, Bella).**

Case 3: Clara → Rocky

1. **Initial Forced Match**: Clara → Rocky.
2. **Final Matching (Children-Proposing Algorithm)**:
   - Alice → Bella
   - Ben → Max
   - Clara → Rocky
   - Daniel → Daisy
   - Emma → Charlie
   - Frank → Luna

Blocking Pair(s):

- **Ben and Rocky**:
  - Ben prefers Rocky (ranked 1st) over Max (2nd).
  - Rocky prefers Ben (ranked 1st) over Clara (2nd).
  - **Blocking pair: (Ben, Rocky).**

The presence of a blocking pair means the children-proposing matching is no longer stable.
Then, the probability of an unstable matching is 5/6 (5 out of 6 potential matches). (3pt)
Explanation ver1: Luna and Clara will always generate blocking pairs if they are not matched to one another. There is only 1 out of 6 cases that Clara will be matched to Luna. (3pt)
Explanation ver2: Show results for the 5 cases with unstable matching (the blocking pair does not need to be (Luna, Clara)). Optional: and the 1 case for stable matching. (3pt)
Steps: Show at least 1 case for an unstable matching and find the blocking pair. (2pt)